

R Programming: Sampling Distributions & Data Simulation

Unit 2: Empirical Study of Sampling Distributions

1. Probability and Probability Distributions

Explanation: Probability distributions describe how probabilities are assigned to different outcomes of a random variable. Common distributions include Normal, Binomial, and Poisson distributions.

Example in R:

```
set.seed(123) # For reproducibility  
random_numbers <- rnorm(5, mean = 0, sd = 1)  
print(random_numbers)
```

2. Generating Random Numbers

Explanation: Generating random numbers is fundamental in simulations and statistical modeling. R provides functions to generate random numbers from various distributions.

Example in R:

```
set.seed(123)  
random_uniform <- runif(5, min = 0, max = 10)  
print(random_uniform)
```

3. Selecting Random Samples

Explanation: Random sampling involves selecting a subset of data from a larger dataset, ensuring each element has an equal chance of being chosen.

Example in R:

```
set.seed(123)  
samples <- sample(1:10, 3)
```

```
print(samples)
```

4. Empirical Study of the Sampling Distribution of Estimators

Explanation: This involves studying how sample statistics, like the mean, vary across different samples from the same population.

Example in R:

```
set.seed(123)
```

```
sample_means <- replicate(1000, mean(rnorm(100, mean = 5, sd = 2)))
```

```
hist(sample_means, main = "Sampling Distribution of the Mean", xlab = "Sample Mean")
```

Unit 3: Data Simulation

1. Simulation of Data from a Probability Distribution

Explanation: Simulating data from a known distribution helps in understanding its properties and in performing statistical analyses.

Example in R:

```
set.seed(123)
```

```
simulated_data <- rnorm(100, mean = 50, sd = 10)
```

2. Simulation of Data for a Regression Model

Explanation: Simulating data for regression involves creating predictor and response variables that follow a specified relationship, often with added noise.

Example in R:

```
set.seed(123)
```

```
x <- rnorm(100, mean = 5, sd = 2)
```

```
epsilon <- rnorm(100, mean = 0, sd = 1)
```

```
y <- 2 * x + epsilon
```

```
lm_model <- lm(y ~ x)
```

```
summary(lm_model)
```

3. Simulation of Data for Time Series Model

Explanation: Simulating time series data involves generating sequences of data points that mimic the behavior of time-dependent processes.

Example in R:

```
set.seed(123)
```

```
ar1_process <- arima.sim(n = 100, list(ar = 0.7))
```

```
plot(ar1_process, main = "Simulated AR(1) Time Series")
```

4. Monte Carlo Simulation

Explanation: Monte Carlo simulations use repeated random sampling to estimate the probability of certain outcomes or to compute integrals.

Example in R:

```
set.seed(123)
```

```
n <- 10000
```

```
x <- runif(n, -1, 1)
```

```
y <- runif(n, -1, 1)
```

```
inside_circle <- sum(x^2 + y^2 <= 1)
```

```
pi_estimate <- 4 * inside_circle / n
```

```
print(pi_estimate)
```

5. Bootstrapping

Explanation: Bootstrapping is a resampling technique used to estimate the distribution of a statistic by repeatedly sampling with replacement from the original data.

Example in R:

```
set.seed(123)
```

```
data <- rnorm(100)
```

```
bootstrap_means <- replicate(1000, mean(sample(data, replace = TRUE)))
```

```
hist(bootstrap_means, main = "Bootstrap Distribution of the Mean", xlab = "Bootstrap Mean")
```

